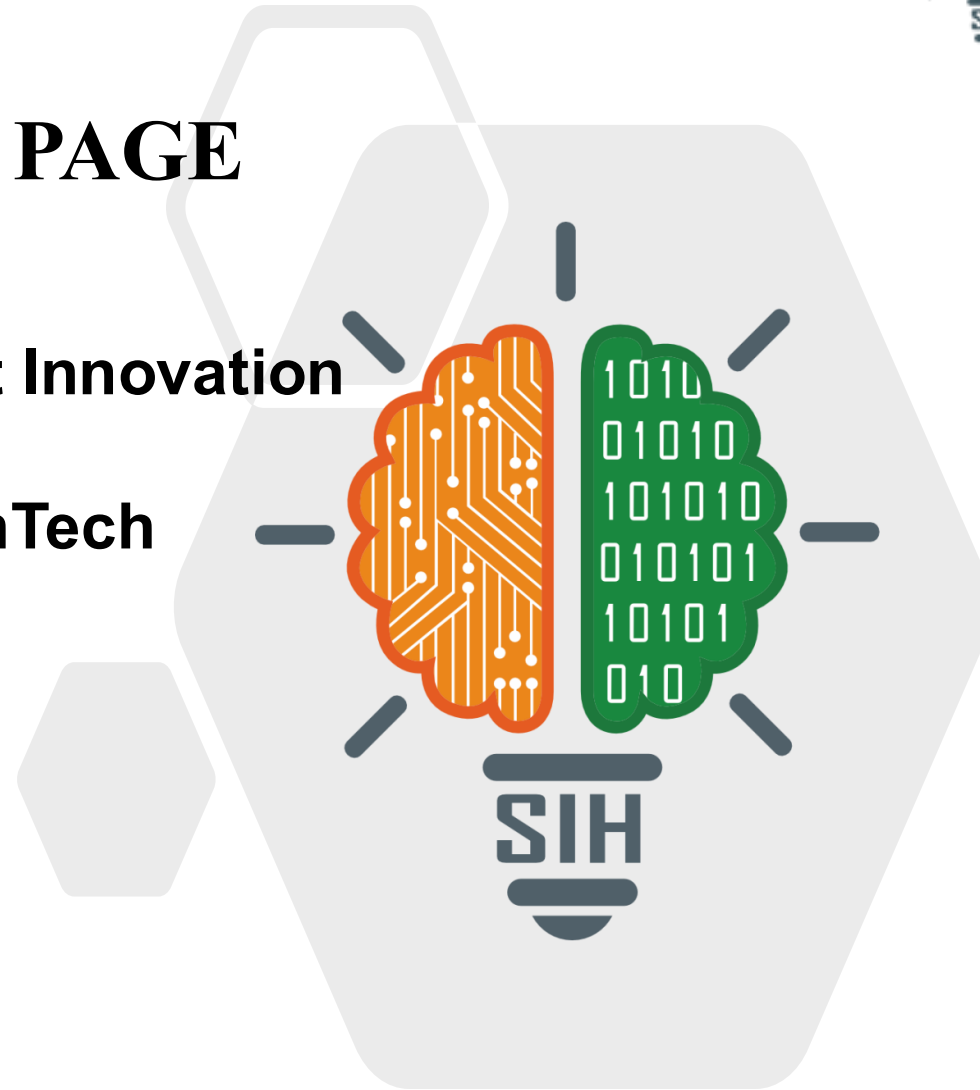


# SMART INDIA HACKATHON 2025



## TITLE PAGE

- Problem Statement ID – SIH25131
- Problem Statement Title- Student Innovation
- Theme- MedTech/ BioTech/ HealthTech
- PS Category- Software
- Team ID-
- Team Name- Runtime Error



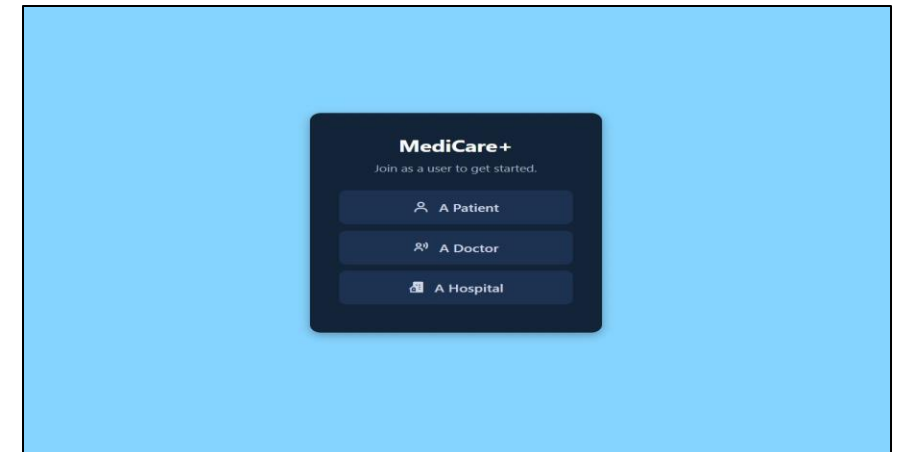
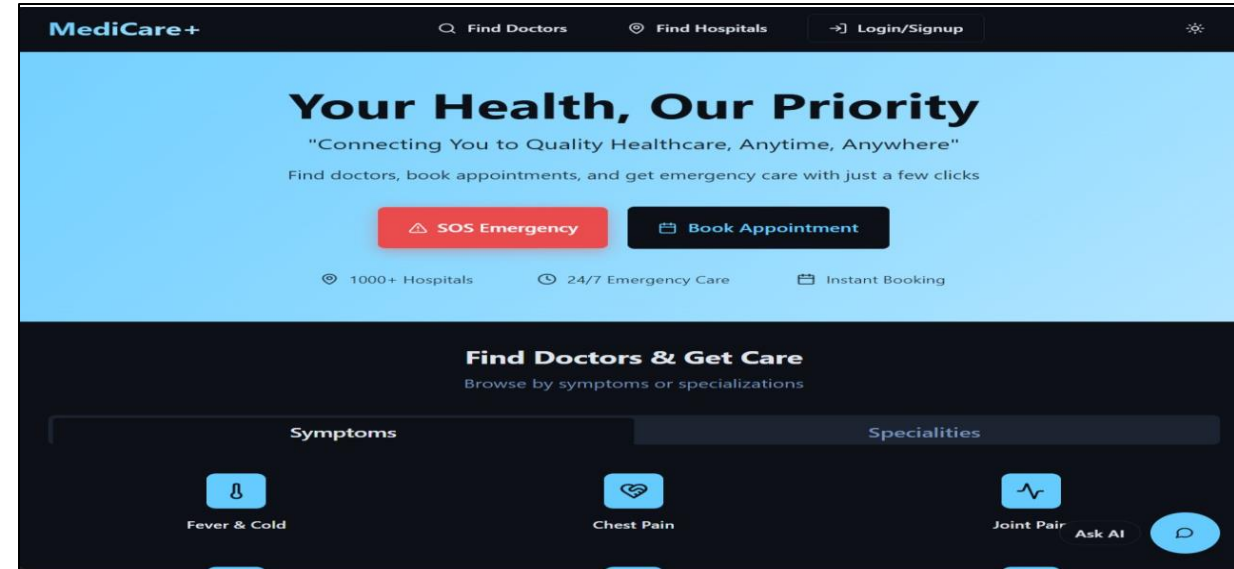
**Problem Statement:** India's healthcare system faces major challenges:

- Fragmented Medical Records
- Long Queues & Appointment Delays
- Incomplete Patient History
- Delayed Lab Reports

**Proposed Solution:** We propose a unified healthcare management platform to simplify access, booking, and records across hospitals.

Our Model Promises:

- Centralized Patient ID & Records
- Online Appointment Booking
- Secure Medical History Access
- Hospital Data & Analytics Dashboard
- Digital Lab Reports Delivery



Runtime  
Error

# TECHNICAL APPROACH

FRONT END :



BACK END:



django

DATABASE :



PostgreSQL

APIs :



django  
REST  
framework

AI & Datascience Libraries:

spaCy



Hugging Face



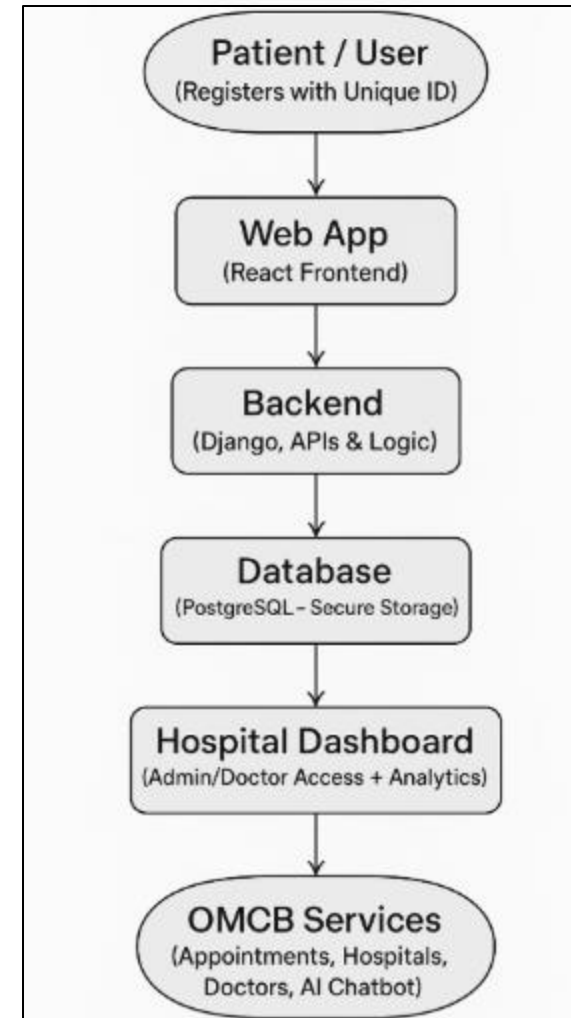
pandas



NumPy



scikit  
learn



# FEASIBILITY AND VIABILITY



## Analysis of the feasibility:

- Rising demand for digital healthcare post-COVID.
- Ayushman Bharat Digital Mission (ABDM) aligns with centralized patient records.
- Direct benefit to patients in rural & tier-2/3 towns (reduce travel, delays).

## Potential Social Challenges:

- Data privacy & security of patient records.
- Integration with diverse hospital systems (manual & digital).
- Regulatory approvals from govt. bodies.
- Scalability: handling millions of patient records efficiently.
- Adoption resistance from hospitals & doctors.

## Strategies to Overcome Challenges:

- End-to-end encryption & role-based access for data safety.
- Pilot programs + government tie-ups for trust & adoption.
- Awareness campaigns in local languages, simple patient UI.
- Cloud-native scalable design for handling big data.

# IMPACT AND BENEFITS



## Potential Impact

- **Patients:** Quick appointment booking, secure health records, personalized alerts & preventive care.
- **Doctors:** Instant access to patient history, smart analytics for better decisions, reduced repetitive tasks.
- **Hospitals:** Streamlined operations, efficient use of staff & beds, improved collaboration across facilities.

## Benefits

- **Societal:** Bridges healthcare gap (urban–rural), raises health awareness, reduces waiting time & overcrowding.
- **Economic:** Cuts patient costs (travel/tests), optimizes hospital resources, supports sustainable public health.
- **Environmental:** Paperless medical records, fewer physical visits → lower carbon footprint.

# RESEARCH AND REFERENCES



- 1) Ayushman Bharat Digital Mission (ABDM), Government of India  
<https://abdm.gov.in/>
- 2) HIPAA (Health Insurance Portability and Accountability Act, USA)  
<https://www.hhs.gov/hipaa/index.html>
- 3) World Health Organization (WHO) – Digital Health Strategy 2020–2025  
<https://www.who.int/publications/i/item/9789240020924>
- 4) NITI Aayog – Health Systems for a New India Report  
[https://www.niti.gov.in/sites/default/files/2019-11/HealthSystemsforaNewIndia\\_0.pdf](https://www.niti.gov.in/sites/default/files/2019-11/HealthSystemsforaNewIndia_0.pdf)